

and hardware setup. In machine learning, construction of algorithms is done through building a mathematical model from input data.

Datasets: Quantity and quality of data is important when it comes to AI applications. Data used to build the final model usually comes from multiple datasets. In most cases, three datasets are used in different stages of the creation of the model. For any given dataset, there are many possible models to identify them with fundamentally different interpretations.

Input data: This is one the most important components when it comes to AI technology. Imagine there are many cameras installed in a retail store having different outlets. From input data received from these camera sensors, the user may have to process thousands of images per minute to detect malicious activities.

Training data: For mission-critical AI applications, accuracy and quality are important. An AI model is useful if input data is trained on. For example, implications of failing to recognise skin cancer can be fatal. We need datasets that are significantly greater in quantity and quality. This means we may have to spend significant time in analysing, preparing and labelling data.

Feedback: A feedback loop improves the model. For example, in the retail store, it would be a good idea to label false predictions and feed them back to the model for retraining. A continuous feedback loop increases training data size, and compute requirements also increase but with better accuracy and results.

Hardware setup: AI is based on algorithms, but without a hardware setup or platform, algorithms do not make any sense. Unfortunately, the importance of hardware is not discussed or included in most of the materials found on the Internet.

An AI ecosystem contains various types of agents including hardware and software. Hardware includes sensors,

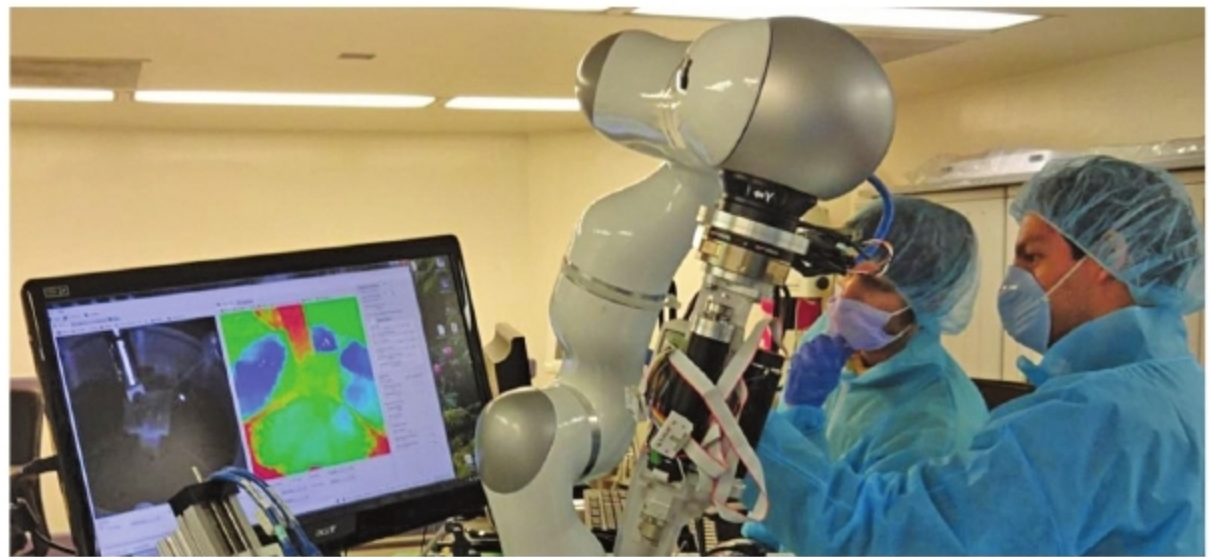


Figure 2: A surgery with the help of an autonomous robot (Credit: <https://blogs.nvidia.com>)

processors and storage. Hardware used for running mission-critical applications also depend on SLAs and differences in data set sizes.

Sensors: These detect or measure physical properties, and record, indicate or otherwise respond to them as per input data. For example, in image processing, input data is fed from vision sensors in a camera system. No matter how complex or good the algorithm is, without a good vision sensor the output will not be accurate as expected.

Processors: The central processing unit (CPU) and graphics processing unit (GPU) are equally important in AI technology. GPU is designed to manipulate and alter memory to accelerate the creation of images. A cluster of GPUs is required to train a model with a training data set consisting of thousands of images. Depending on data size and performance of the model, the number of input images and SLAs expected, a cluster of processors are used to meet application requirements.

Storage: To train AI and machine learning algorithms, large datasets are needed to deliver accurate decisions. This means that the demand for data storage space is significant. Managing data sets requires storage systems that can scale without limits. Your data set may not fit on the disk of a laptop, and you may need a number of SSDs and/or HDDs to store the ever-increasing training data.

AI helps make better and smarter decisions

AI technology is employed in safety-critical situations such as airports, ATM machines and aircraft operations. When decision-makers and business executives have reliable data analyses, recommendations and follow-ups through AI systems, they can make smarter decisions and better choices for their business, employees and future actions.

Automotive: In the automotive industry, several AI applications have been developed, from vehicle design to marketing and sales decision-making support. For instance, AI has led to the design of smarter cars, equipped with multiple sensors that learn and identify patterns, and add-on safe-drive features that warn drivers of possible collisions and lane departures. AI is used to map the decisions made by car buyers and manufacturers, and transportation services providers.

Self-driving cars: AI takes decisions by itself in a self-driving car where computer vision and machine learning plays a vital role. There are a number of interesting and challenging decision-making problems, starting from autonomous driving of a single vehicle to coordination among multiple vehicles. AI algorithms take care of these problems through an autonomous system throughout the course of driving till the destination is reached.